Unit: 6 Unit 5: Linear Equations
Topic: Rate of Change and Slope

Name: ________________
Date: __________

Foundation (Jalapeno)

Q1. What is the slope of the line $y = 2x + 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. A line has a slope of $\frac{1}{2}$. What is the rate of change?
(A) 1
(B) $\frac{1}{2}$
(C) 2
(D) 4
(E) 6

Q3. What is the slope of a horizontal line?
(A) 0
(B) 1
(C) 2
(D) undefined
(E) none

Q4. What type of slope does a vertical line have?
(A) 0
(B) 1
(C) 2
(D) undefined
(E) none

Q5. A line passes through points $(2, 3)$ and $(4, 5)$. What is the slope?
(A) 0
(B) 1
(C) $\frac{1}{2}$
(D) $\frac{2}{2}$
(E) $\frac{3}{2}$

Q6. The table shows values of x and y. What is the slope of the line?
(A) $\frac{1}{2}$
(B) 1
(C) 2
(D) 3
(E) 4

Q7. Given $y = -3x + 2$. Is the slope positive or negative?
(A) positive
(B) negative
(C) zero
(D) undefined
(E) none

Q8. Which equation represents a line with a slope of -2 and a y-intercept of 3?
(A) $y = 2x + 3$
(B) $y = -2x + 3$
(C) $y = 3x - 2$
(D) $y = -3x + 2$
(E) $y = 2x - 3$

Open Ended Questions (FRQ)

Q9. Find the slope of the line that passes through the points $(1, 2)$ and $(3, 5)$ and explain what the slope represents.

Q10. The cost of renting a car is \$40 per day plus a one-time fee of \$20. Write an equation to represent the cost y in terms of the number of days x rented. Then, find the slope and interpret its meaning.

Chef's Tips

- Emphasize the concept of rate of change.
- Use real-life examples.
- Encourage graphing.